

Homework 3: Logistic regression and GAMs

ENGM 182 Winter 2026

Due at 11:59 pm ET on Monday, 9 February 2026 via Canvas

GRADING AND GENERAL INSTRUCTIONS: The problems add up to 90 points. The remaining 10 points are reserved for style and clarity. Please submit a Python Notebook or an R Markdown file. Late assignments will be scored in accordance with the policy in the course syllabus.

In this homework, we will be using a dataset, based on digitalized images of cell nuclei in tissue, to diagnose whether a breast tumor is malignant or benign. The dataset (`breast_cancer.csv`) is comprised of 569 observations. The response variable, `diagnosis`, indicates whether a tumor is malignant (M) or benign (B). There are 30 independent variables related to attributes of the cell nuclei (including radius, texture, perimeter, area, smoothness, compactness, concavity, concavity point, symmetry, and fractal dimension). The mean, error, and worst values for these attributes are provided.

1. (5) Convert the response variable `diagnosis` to a binary representation, with 1 as malignant.
2. (25) Fit a logistic regression model to predict the probability that a tumor is malignant. Include the independent variables partially chosen for low multicollinearity: `"mean_perimeter", "mean_symmetry", "worst_radius", "worst_texture", "worst_concavity_points", "worst_symmetry"`. Report the coefficients, standard errors, and p-values. Provide a conceptual interpretation of the results, including how the attributes of the cell nuclei influence the probability of malignancy.
3. (20) Split the data into a training and testing set (90-10 split), using the same explanatory variables as above. Train a regularized logistic regression model with regularization parameter $\lambda=1$. Plot a confusion matrix to evaluate the predictions of the model on the testing set. How would we go about choosing a best performing λ in practice?
4. (20) Fit a logistic Generalized Additive Model to the same training data/features from Problem 3, and select regularization parameters from at least $\lambda = 1, 1000, \text{ and } 10000$ with best estimated out of sample performance (**hint: pyGAM's gridsearch function is likely much easier and faster than using full cross validation. The process may take a couple minutes**). What are the best performing regularization parameters? How do you interpret this? Are any feature functions close to linear?
5. (10) Plot partial dependence functions for all explanatory variables, which show the estimated effect of an explanatory variable on a response for varied values of that parameter (**hint: you can copy the code almost exactly from the Generalized Additive Models in Python.ipynb Notebook, potentially changing the model name**). Give your interpretation of the predicted effects of the explanatory variables - how do they compare to the estimated logistic regression coefficients?
6. (10) Make predictions using your GAM model and create a confusion matrix to evaluate model predictive performance on the test set. (**Hint: since pyGAM is not a scikit learn model, you will want to generate your predictions `y_pred` and plot the confusion**

matrix from those and your true y values, rather than using the `ConfusionMatrixDisplay.from_estimator` scikit learn function. You can do this as `ConfusionMatrixDisplay.from_predictions(y_test, y_pred)`). How do the results compare to the logistic regression case?